Bartonella Neuroretinitis Following Dog Bite: an Uncommon Case of Non-Feline Related Cat Scratch Disease and Review of Literature

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Short Report

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Abstract

Background: Neuroretinitis is the most common form of ocular Cat Scratch Disease (CSD), caused by bacterium Bartonella Henselae, commonly transmitted to human via cat's scratch or bite. We describe a case of Bartonella neuroretinitis with occlusive vasculitis without history of feline exposure.

Case Presentation: A healthy 28 year-old man presented with unilateral acute visual loss following a dog bite and diagnosed as bartonella neuroretinitis with occlusive vasculitis, with subsequent recurrence. Retinal imaging included colour fundal photography and optical coherence tomography (OCT). Patient improved upon commencement of oral azithromycin combined with prednisolone. Our literature review revealed 9 cases of serology confirmed Bartonella-associated neuroretinitis following direct contact with non-feline mammals (dog, guinea pig, ferret, hedgehog, raccoon), tick and bull ant.

Conclusion: Due to its variable mode of transmission, the diagnosis of Cat Scratch Disease and Bartonella-associated neuroretinitis should be considered despite absence of feline exposure.

1. Introduction

Cat Scratch Disease (CSD) is a self-limiting systemic infectious disease caused by Bartonella Henselae (B. Henselae), a gram negative bacterium classically transmitted to human with history of cat's exposure via scratches, licks or bites. Incidence of B.Henselae seropositivity rate is quoted as 3% in the general population[1], but only a minority of exposures to B. Henselae in immunocompetent individuals resulted in CSD and it is with excellent prognosis. However, the name Cat Scratch Disease may serve as a misnomer as not only cats carry B. Hensalae. There are increasing number of reported CSD cases without any feline exposure.

The hallmark of systemic manifestation of CSD is fever, with the presence of inoculation papule and regional lymphadenopathy involving the draining sites of inoculation. Ocular involvement occurs in only 2–4% of patients[3,6] with variable manifestations. Neuroretinitis is the most common form of ocular CSD[2,4,7], apart from Parinaud's oculoglandular syndrome, multifocal retinitis, uveitis and retinal artery occlusion[2,6] ,. On the contrary, B. Henselae has been observed as among the most common cause of neuroretinitis[5,39] .

2. Case Presentation

A healthy 28 year old man was referred to Ophthalmology Department, University Malaya Medical Centre with sudden onset of blurring of vision in his right eye of 1 week duration, associated with flu-like symptoms. He reported a history of dog bites 2 weeks prior. Visual acuity was 6/60 on his right eye, with no improvement upon correction, and 6/6 unaided is his left eye. Optic nerve function was impaired in the right eye, with presence of relative afferent pupillary defect, dyschromatopsia, central scotoma and inferior hemifield defect on visual field test, reduced red colour saturation and light brightness. Slit lamp examination revealed cells 2 + in anterior chamber with fine keratic precipitates inferiorly. Intraocular pressure (IOP) was 12 and 14 mmHg in right and left eye, respectively. Dilated funduscopy (Fig. 1) showed swollen optic disc, with multiple peripapillary flame-shaped haemorrhages, cotton wool spots, periphlebitis with marked retinal vessels sheathing superior and inferior to the disc surrounded by intraretinal haemorrhages, suggestive of occlusive vasculitis. There was no retinitis or choroiditis. OCT macula (Fig. 2) revealed serous macula oedema with central retinal thickness of 472 um. The left eye was normal and completely uninvolved. Topical steroid and cycloplegic eye drops were commenced for the right eye.

Five days later, right eye worsened with best corrected vision of 2/60, vitritis and macular striae developing, alongside optic disc swelling and superotemporal and less marked inferotemporal retinal vessels sheathing. Left eye began to show similar vessels sheathing in one of the inferior retinal vessels with no choroiditis, retinitis or vitritis. Left visual acuity maintained at 6/6.

Investigations:

Serology for B. Henselae was positive with titre of 128 units. Work up for other infectious causes of his vitritis were negative, including toxoplasma, syphilis, tuberculosis, leptospirosis and HIV. A complete blood count with differential, complete metabolic panel, serum ACE, ANA, ANCA, rheumatoid factor and chest X-ray were also normal. Fundal fluorescein angiogram of right eye revealed enhanced retinal vessels, masking effect in area superior to optic disc (from hemorrhages), vasculitis involving both retinal arteries and veins and a large area of capillary non-perfusion superotemporally. Left eye showed inferior vessel enhancement suggestive of vasculitis.

Treatment:

Intravenous methylprednisolone 500mg twice daily commenced for three days followed by 1mg/kg of oral prednisolone in tapering dose. Oral Azithromycin was initiated at 500 mg orally once per day, in addition to the patient's present topical regimen of steroids and IOP-lowering medications. Once inflammation subsided, segmental retinal photocoagulation was done to the corresponding area of non-perfusion on fundal fluorescein angiogram.

Outcome:

Vitritis persisted following treatment with steroid alone. Fortunately after one week of treatment with azithromycin and steroid, there was gradual recovery of visual acuity to best corrected of 6/36 with resolution of vitritis and optic disc swelling, along with reduction in macular oedema. A month later (Fig. 3), right eye best corrected visual acuity maintained at 6/36 with normal intraocular pressure. Visual Evoked Potential done revealed a prolonged P100 latency of the right eye, which signifies a conduction defect on the optic nerve. Azithromycin was discontinued after a month course and oral steroid was tapered down over 6 weeks.

Recurrence :

9 months following the first episode, the patient presented with 4 days history of progressive blurring of vision of his right eye. Right eye visual acuity deteriorated to 3/60 with relative afferent pupillary defect as previously documented, and left eye visual acuity maintained at 6/6. Anterior chamber was deep with cells + 2. Intraocular pressure was 12 mmHg in both eyes. Funduscopy of right eye (Fig. 4) revealed optic disc swelling, perivascular sheathing with retinal haemorrhages on all quadrants. There was no vitritis, retinitis or choroiditis. OCT macula (Fig. 5) showed loss of foveal contour, macular oedema and epiretinal membrane. Oral Azithromycin 500mg once daily was commenced alongside 1mg/kg of oral prednisolone, topical steroid and cycloplegic eyedrops. Gradual improvement of optic disc swelling and vessels sheathing was observed within 6 weeks and right eye best corrected visual acuity returned to baseline, 6/36.

3. Discussion

Neuroretinitis is defined as a type of optic neuropathy characterized by an acute visual loss with optic disc swelling, accompanied by hard exudates characteristically arranged in a star shape around the fovea. We described a case of ocular CSD in its most common form, which is neuroretinitis. There was no feline exposure, but there was a dog bite instead. Cats are well known to be the primary host for Bartonella Henselae, in which the infected cats have asymptomatic bacteraemia lasting for months[7]. However, the name Cat Scratch Disease may serve as a misnomer as not only cats carry B.Hensalae. Fleas, ticks and lices has been identified as vectors for the transmission of B.Hensalae from feline to other animals[8]. As B.Hensalae, the causative microorganism of CSD is difficult to culture, clinical diagnosis of CSD is most often made following history of cat exposure, physical examination and confirmed with serologic test with high titres of immunoglobulin G to B. Henselae. Although majority of CSD patients has history of contact with cat, there has been increasing documented cases of ocular CSD without prior contact with cat or known vectors[10,11,12].

Serology confirmed Bartonella-associated neuroretinitis has been reported following contact with other domestic animals. Our literature search yielded 9 articles with a total of 17 eyes that contracted neuroretinitis without feline exposure (Table 1). Including our patient, 6 out of the 13 (46%) patients had recent history of contact with dog prior to onset of the symptoms, while the others involved bites from guinea pig, ferret, raccoon, bull ant and tick, as well as exposure to hedgehog. These recent findings implicate that dogs and other non-feline domestic animals can be the reservoir of B. Henselae, with blood-sucking arthropods as vectors. It is known that Bartonella species has excellent specific adaptation strategies to its preferred hosts as well as survival in the vector feces[13] .

It is also important to note that cross reactivity in serology testing has been observed between different Bartonella species; B. Henselae with B.Grahamii[14] as well as B. Quintana[15,16], both of which were reported to cause neuroretinitis. Interestingly, a case of B. Elizabethae neuroretinitis was also reported following raccoon bite[17] .

Prodromal illness of fever, malaise and headache is only present in 7 out of 13 patients, while lymphadenopathy is even less. Our patient's systemic examination did not reveal any lymphadenopathy, which is the hallmark of systemic CSD, as it only occurs in 20% of ocular CSD cases[18,19]. In spite of this, the finding of regional lymphadenopathy in neuroretinitis patients serves as an important clue for the diagnosis of CSD.

Bartonella-associated neuroretinitis may be unilateral or bilateral in either immunocompromised or immunocompetent patient[19]. Our patient presented with the typical unilateral neuroretinitis with right eye optic disc swelling, optic nerve dysfunction, subretinal fluid at macula with macula star emerging two weeks later. Habot-Wilner et al reported 90% of ocular CSD patients had optic disc swelling upon presentation, whereby 80% of this is due to neuroretinitis and the rest is diagnosed as inflammatory disc oedema[18].

Apart from neuroretinitis, our patient showed presence of vessels sheathing, with involvement of veins more than arteries. He had occlusive vasculitis with a large area of capillary non-perfusion superotemporally. Around 7% of reported ocular CSD cases manifested as retinal vessel occlusions[18,20,21], where it was observed that retinal arteries were more commonly involved than veins[18,20]. Presence of retinal vasculitis is even less frequently reported[21]. However, this is not a surprise as Bartonella has been shown to interact and invade endothelium, thus may elicit obliterative vasculitis[23]. Thus, it is also reasonable to consider CSD in cases of retinal vessel occlusion in patients lacking of known vascular risk factors.

The visual evoked potential (VEP) of our patient at two months following his initial presentation was abnormal, in which his right eye P100 latency was prolonged, signifying a conduction defect in the optic nerve. His optic nerve function remained impaired, as expected that some degree of optic neuropathy will remain even after resolution of neuroretinitis[19]. This can be explained by the pathogenesis of neuroretinitis as it is associated with direct involvement of the optic nerve head fibers as the primary target[24] by infectious or inflammatory processes, leading to edema and fluid exudation from the inflamed cellular area of the peripapillary retina[25]. Therefore in bartonella neuroretinitis, the VEP will be abnormal with prolongation of latency of P100 wave and a decrease in the amplitude. However, electroretinogram (ERG) will be normal[8], as the functional integrity of the retinal layers are not disrupted.

As a self-limiting disorder, the treatment of ocular cat scratch disease has been slightly controversial. Reed et al concluded that the use doxycycline and rifampicin shorten course of disease and hasten visual recovery[19]. This is consistent with recommendation from Fairbanks et al for a four to six weeks regimen of doxycycline or azithromycin alongside rifampicin in patients who suffer severe vision loss and/or moderate to severe systemic symptoms[26]. However Solley et al found no difference in the final visual outcome between treated and untreated ocular CSD[20]. Habot-Wilner suggested following closely patients with ocular CSD that retained their good vision upon presentation, without initiation of systemic antibiotic. However they recommended a combination of systemic corticosteroid and antibiotic for moderate to severe visual loss[18]. The use of corticosteroid therapy remains constroversial in infectious neuroretinitis cases. Chrousus et al reported systemic corticosteroids with or without systemic antibiotics to be effective in this condition[27]. The use of azithromycin has been shown to be efficacious in a prospective randomized placebo-controlled trial by Bass et al[27] and Ives et al[28]. Both our patients showed dramatic recovery upon commencement of azithromycin. Other options would include erythromycin, doxycycline, rifampicin, ciprofloxacin and aminoglycosides[30] .

4. Conclusions

Due to its variable mode of transmission, the diagnosis of Cat Scratch Disease and Bartonella-associated neuroretinitis should be considered despite absence of feline exposure.

Abbreviations

CSD
cat scratch disease
OCT
optical coherence tomography
IOP
intraocular pressure
ACE
angiotensin coverting enzyme
ANA
antinuclear antibody
ANCA
anti-neutrophil cytoplasmic antibody
VEP
visual evoked potential
ERG
electroretinogram

Declarations

Ethics approval and consent to participate: not applicable

Consent for publication :
Informed audiovisual consent was obtained from the subject.

Availability of data and material :

All data generated or analysed during this study are included in this published article.

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Authors contributions :
F.R. provided the acquisition of data, analysis and interpretation of data, drafting the article, revised it critically for important intellectual content, and final approval of the version to be submitted; I.T. provided the data interpretation, revised the article critically for important intellectual content and gave final approval of the version to be submitted; T.A.K. and P.W.L. was responsible for the article critically for important intellectual content.

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Tables

Table 1

	Year	No	Age	Ocular	Micro organism	Eye	Lymphadenopathy	Systemic symptom	Nature of animal exposure	IFA IgG titres	Treatment	P V
1	**Hernandez-Da 2009**	1	7	Neuroretinitis, choroiditis	B.Henselae	BE	no	nil	Dog scratch	ELISA	Corticosteroid Erythromycin	5
2	**Uchi 2019**	1	24	Neuroretinitis	B.Henselae	RE	missing	Fever malaise	Contact with dog	1:40 IgM	Nil (patient was pregnant)	6
3	**Kerkhoff 1999**	3	62	Panuveitis	B.Henselae	BE	yes	fever	Contact with dog	1:2000	Rifampicin Doxycycline	R L
			65	Neuroretinitis	B.Henselae	LE	nil	nil	Contact with dog	1:1800	Ciprofloxacin	C
			45	Panuveitis	B.Henselae	BE	nil	nil	Contact with dog	1:250 IgM	Rifampicin Doxycycline	R L
4	**Orellana-Rios 2018**	2	51	RE neuroretinitis LE retinitis and vitritis	B.Henselae	BE	yes	Fever headache	Ferret bite	1:512	Azithromycin	R L
			19	RE retinitis LE neuroretinitis	B.Henselae	BE	yes	Fever	Guinea pig bite	1:1024	Rifampicin Doxycycline	R L
5	**Kiu 2015**	1	22	Neuroretinitis, DUSN	B.Henselae	LE	no	nil	Contact with hedgehog and guinea pig	1:48 IgM	Doxycycline Albendazole	6
6	**OHalloran 1998**	1	30	Neuroretinitis	B.Elizabethae	RE	no	nil	Raccoon bite	1:128	Erythromycin	C
7	**Ullrich 2012**	1	52	Neuroretinitis	B.Henselae	RE	no	rigors	Bull ant bite	1:256	Erythromycin	H
8	**Kalogeropoulos 2018**	1	29	Vitritis	B.Quintana	RE	no	nil	Tick/flea bite	1:256	Rifampicin Doxycycline	6
9	**Anders 2015**	1	52	Neuroretinitis Serous choroidal detachment	B.Henselae	LE	no	Fever headache	Tick bite	Raised	Corticosteroid Doxycycline	C

Figures

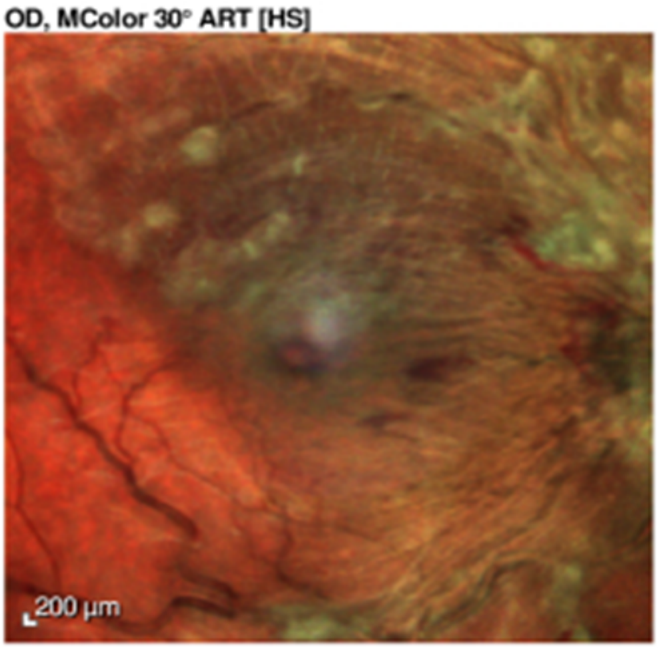

Figure 1

Right eye swollen optic disc with peripapillary flame-shaped haemorrhages and retinal vessels sheathing

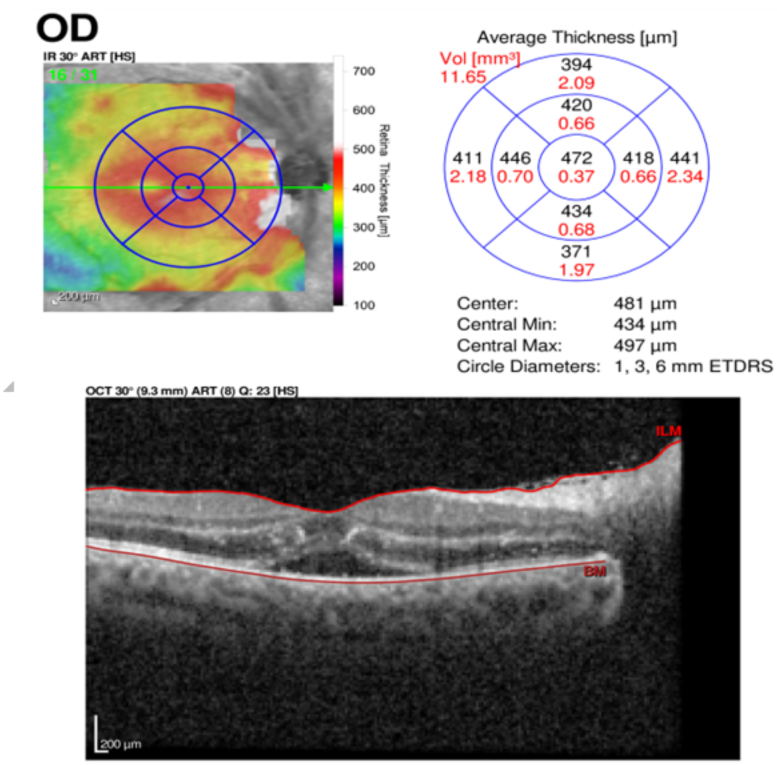

Figure 2

OCT macula right eye showed subfoveal subretinal fluid.

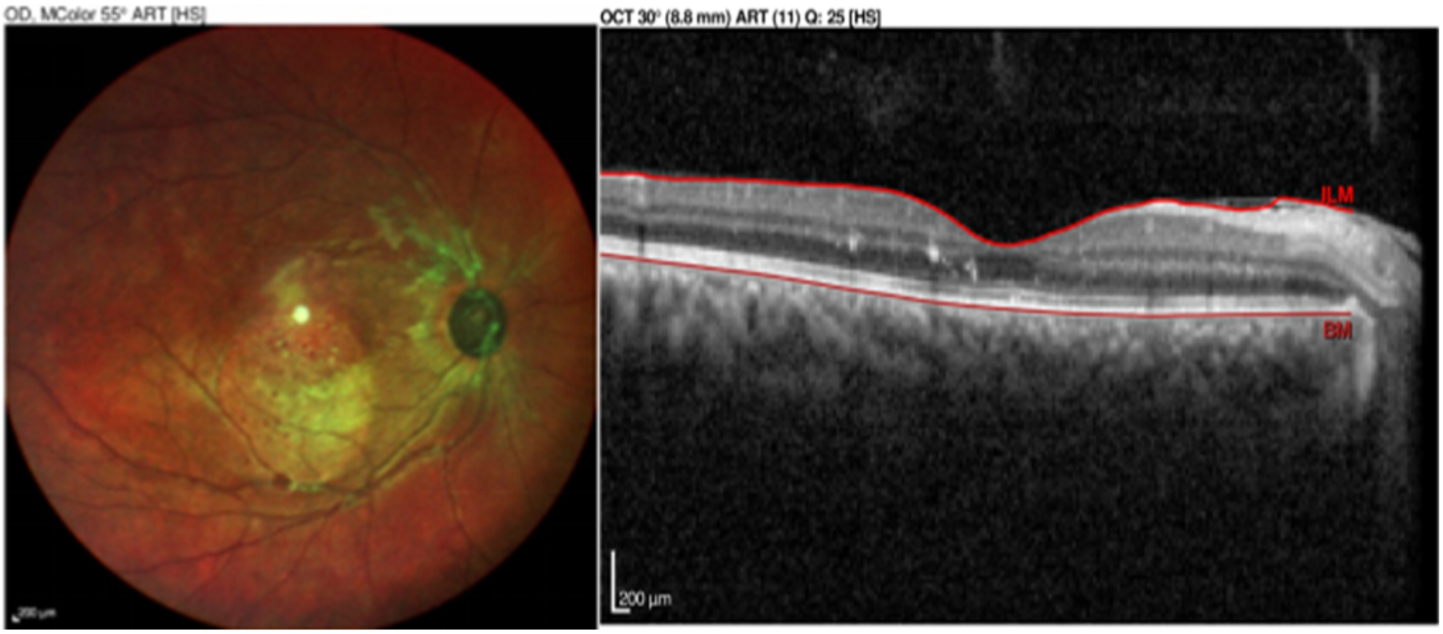

Figure 3

Right eye reduced optic disc swelling and vessels sheathing. OCT macula showed resolved macular oedema.

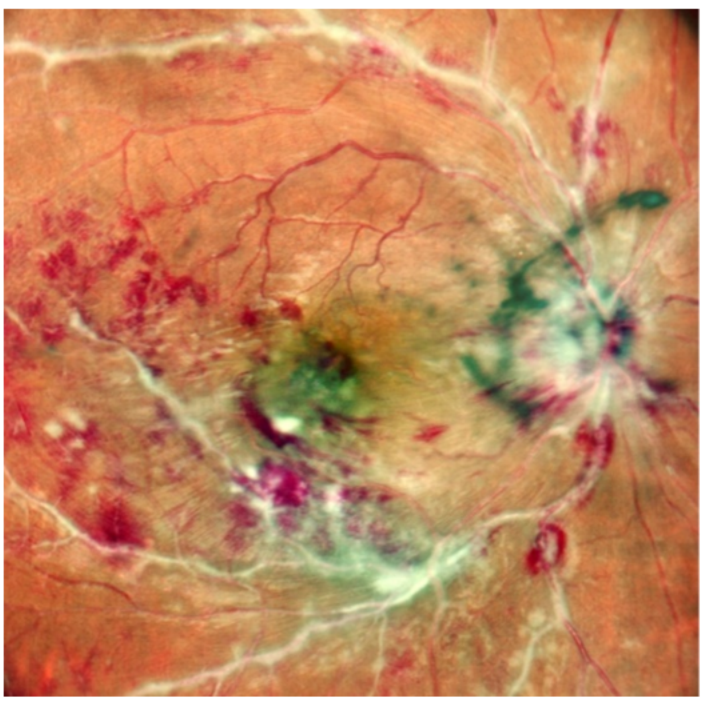

Figure 4

Right eye optic disc swelling, perivascular sheathing and retinal haemorrhages.

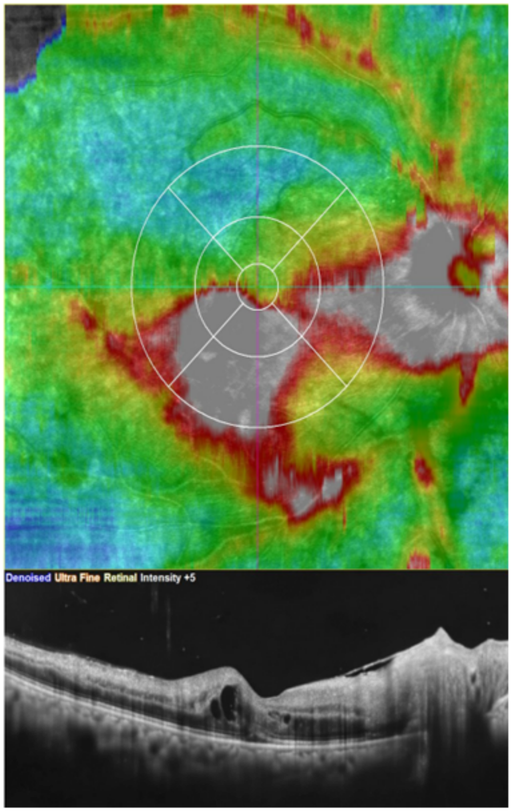

Figure 5

OCT macula showed intraretinal and subretinal fluid with epiretinal membrane.